University of Manitoba
Faculty of Science
Department of Mathematics

Contents

1 Course Details

Course Title & Number	MATH 1500: Introduction to calculus
Number of Credit Hours	3
Term	Fall 2023
Class attendance	Mandatory

2 Instructor Contact Information

Instructor Name	Vitalii Akimenko (A01)
Office Hours	TBD
Email	Vitalii.Akimenko@umanitoba.ca

Instructor Name	Xinli Wang (A02)
Office Hours or Availability	In-person: Wed 11:00pm –12:00pm Machray Hall #427 Online by appointment: Teams Booking
Email	xinli.wang@umanitoba.ca

Instructor Name	Liliana Menjivar Lopez (A03)
Office Location	452 Machray Hall
Office Hours	TBD (in person or via Cisco Webex on UMLearn).
Email	Liliana.MenjivarLopez@umanitoba.ca

Instructor Name	Darja Barr (A05)
Office Location	436 Machray Hall
Office Hours	Online and in person available, times will vary - email for appointments
Email	Darja.Barr@umanitoba.ca

Instructor Name	Craig Cowan (A06 and A07)
Office Hours	TBD (in person or via Cisco Webex on UMLearn) 435 Machray Hall
Email	craig.cowan@umanitoba.ca

Instructor Name	C. Wang (A08)
Office Hours	Mondays 10:30 - 11:20am in person in Machary Hall 433 Thursdays 9-9:50am on Zoom Click here to join
Email	caelan.wang@umanitoba.ca When emailing me, please use your University of Manitoba email and include ”MATH 1500 A08” in the title. I aim to respond to all my emails within 48 hours, excluding evenings, weekends, and holidays.

3 Class Schedule

This schedule is subject to change at the discretion of the teaching team and/or based on the learning needs of you. Changes are subject to Section 2.8 of the ROASS Procedure.

Schedule Table:

Preliminary Schedule

Week	Topics	Assessment
1(Sep 06– 08)	1.1 Reviews of functions 1.2 Basic classes of functions	
2(Sep 11–15)	1.4 Inverse functions 1.5 Exponential and Logarithmic Functions (exclude hyperbolic functions) 2.2 The limit of a function	Diagnostic Test
3(Sep 18–22)	2.3 The limit laws	
4(Sep 25-29)	2.4 Continuity 2.6 Limits at Infinity and Asymptotes	
5(Oct 02–06)	3.1 Defining the Derivative 3.4 Derivatives as Rates of Change (Velocity/ Acceleration only) 3.2 The Derivative as a Function	Quiz 1
6(Oct 09–13)	3.3 Differentiation Rules	
7(Oct 16-20)	3.5 Derivatives of Trigonometric Functions 3.6 The Chain Rule	Quiz 2
8(Oct 23-27)	3.8 Implicit Differentiation 3.9 Derivatives of Exponential and Logarithmic Functions	
9(Oct 30 –Nov 03)	4.1 Related Rates 4.3 Maxima and Minima	Nov 02 Mid-Term 5:45 – 6:45pm
10(Nov 06–10)	4.4 The Mean Value Theorem 4.5 Derivatives and the Shape of a Graph	
11(Nov 20-24)	4.6 Curve Sketching (omit oblique asymptotes) 4.7 Applied Optimization Problems	Quiz 3
12(Nov 27-Dec 01)	4.10 Antiderivatives 5.1 Approximating Areas	
13(Dec 04–Dec 08)	5.2 The Definite Integral 5.3 The Fundamental Theorem of Calculus	Quiz 4
14(Dec 11)	Review	

Quizzes always happen during lab/tutorial time. All topics taught in the course, including proofs, are testable. No calculational aids of any kind (calculator, smartphone, abacus) are permitted on the diagnostic test, quizzes, the midterm, and the final exam. Re-weighting of final grade or make-ups will not be considered for poor performance on any assessment.

Important dates

- Sep 19, 2023: Last date to drop Fall term and Fall/Winter term spanning courses with refunds
- Oct 02, 2023: National Day for Truth and Reconciliation observed - University is closed
- Oct 09, 2023: Thanksgiving Day - University is closed
- Nov 13, 2023 - Nov. 17, 2023: Fall Term Break
- Nov 21, 2023: Voluntary withdrawal deadline for Fall courses.

4 Textbook, Readings, Materials

Required textbook: “Calculus Volume 1” by Gilbert Strang and Edwin ”Jed” Herman. This textbook is an adaptation of Open Education Resource on OpenStax. The adaptations were completed by Dr. Nicholas Harland.

You can read the whole book online under the license CC BY NC SA. PDF versions and other formats will be available on UMLearn and may be printed by the student.

Calculus by James Stewart may be used as an optional resource.

The bookstore also sells a ”course pack” of prior MATH 1500 Midterm and Final Exams, with solutions, to help students prepare for exams. This course pack is not required. It is up to you whether you purchase it!

5 Course Evaluation Methods

# of assessment(s)	Assessment	Value of Final Grade
1	Diagnostic Test	5%
12	Knewton Alta Online Assignment	1% each best 11/12 count 11% in total
1	Mid-Term	20%
4	Quiz	7% each 28% in total
1	Final Exam	36%

If you miss one quiz, 7% will be moved to your final exam.

If you miss two quizzes, 14% will be moved to your final exam.

Any student who misses at least 3 quizzes cannot achieve a grade higher than an F in this course.

Make-up quizzes will not be provided for any reason. While no doctor's notes are required, students who miss writing a quiz must contact and notify the course coordinator, Dr. Xinli Wang no later than 48 hours after their scheduled quiz (but preferably earlier) for record keeping and for guidance. You will need to fill up a self-declaration form and include it in your email to Dr. Wang.

Do not miss the mid-term exam. It's compulsory for everyone to take it. If students are unable to attend the midterm for a valid reason, they must contact their instructor within 48 hours of missing the test to request a deferral.

You will need to get at least 50% for final exam in order to pass the course.

6 Grading

Letter Grade	Minimum Percentage to Guarantee	Final Grade Point
A+	91	4.5
A	83	4.0
B+	78	3.5
B	72	3.0
C+	66	2.5
C	60	2.0
D	51	1.0
F	Less than 51	0

7 Quizzes and Mid-Term Grading Times

It is expected that quizzes and mid-term will be marked and returned within two weeks.

8 Description of assessments

8.1 Diagnostic Test

Step A: The First Pre-Calculus Diagnostic Test

The diagnostic test is run online in UMLearn. The first chance at the diagnostic test will occur online in UMLearn from **Friday Sep 15th at 9:00am until Monday Sep 18th 4:00pm. All students are expected to write this test before the due date / time.**

- If you obtain a grade of 78% or higher on the first diagnostic test, then you get the whole 5% of your diagnostic test grade and you are done (we will call this Group 1).
- We will call Group 2 those who obtain between 55% and below 78% on the first diagnostic test, and Group 3 those below 55%. Students who do not take the first diagnostic test for any reason during the time allotted (for example you do not write it in the time given or you registered late for the course) are also in Group 3.
- If you are in Groups 2 or 3, go to Step B.
- If you join the class late, you will need to go to Step B by default.

Step B: Remediation

- If your grade is less than 55% on the first diagnostic test or if you did not write the first attempt of the diagnostic test, you are required to complete Remediation Module 1 (Exponents, Solving by Factoring, Rational Expressions) and Module 2 (Functions, Logarithms and Trigonometry) on Knewton Alta.
- If your grade is at least 55% but less than 78%, you are required to complete the Remediation Module 2 (Functions, Logarithms and Trigonometry) on Knewton Alta.
- Note: a module will be considered “complete” only if you have achieved 100% mastery for all objectives from that module. To be eligible to re-write the diagnostic test (Step 3 below), you must complete the required remediation as outlined above. Failure to complete the required remediation results in an automatic grade of 0 for the diagnostic test.

REMEDIATION MUST BE COMPLETED BY Oct 4th, 2023. However it is strongly recommended that you complete the remediation as soon as possible. Upon completion of the required remediation, please go to Step C.

Step C: The Second Diagnostic Test

- Those in Groups 2 and 3 are required to write diagnostic test #2 online between **Thursday Oct 5th at 11am until Sunday Oct 8th at 4:00pm**.

How Your Diagnostic Mark Will Be Determined

- If you are in Group 1 (that is, you obtained a grade of 78% or higher on the first diagnostic test), then you get the whole 5% of your diagnostic test grade and you are done.
- If you are in Group 2 or 3 (that is, you obtained a grade of below 78% on the first diagnostic test or you didn't write it), and you do not complete the required remediation within KnewtonAlta, **your mark for the diagnostic test will be 0, regardless of your grade on the second diagnostic test**.
- If you are in Group 2 (that is, you obtained a grade of below 78% on the first diagnostic test), and you get at least 78% on the second diagnostic test, you will get the full 5 marks. Otherwise you get your percentage on the second diagnostic test of the 5 marks.
- If you are in Group 3 (that is, you obtained a grade of less than 55% on the first diagnostic test), the percent of the 5 marks you will get is **the LOWEST PERCENTAGE** you achieved on each of Remediation Module 1, Remediation Module 2, and on the second diagnostic test.

8.2 Knewton Alta Online Assigment

These weekly online assignments help you master core topics in this course. Note these assignments are mastery-based. In other words, you won't be able to complete them till you have shown mastery of a certain topic. Each two weeks there are two sets due on Sunday 11:59pm. Your mastery percentage on an assignment will be your mark on that assignment. Please purchase a code from the campus bookstore in order to access the platform. More details can be found on UM Learn.

Late submission of the assignments are accepted, but only if 100% mastery is achieved, and it comes with a 10% per day penalty. It may take Knewton Alta up to 48 hours to update the marks of late submissions, so please do wait for a couple of days if you don't see an update of your mark right away.

Here are a few examples for you to better understand how late submissions work.

1. Suppose you had 0% mastery by the time an assignment is due. If you achieved 100% mastery on the second day after an assignment is due, you will receive 100%-20%=80% for that assignment.
2. Suppose you have achieved 60% mastery on an assignment by the due date, and then eventually achieved 100% on the 7th day after the due date. Note 100%-70%=30%, which is less than 60%. So you would receive the 60% as it is higher.
3. Suppose you have achieved 50% mastery on an assignment by the due date, and then worked on it a little bit more and achieved 90% eventually. Since you did not achieve 100%, the late submission is not considered complete, and therefore the original mark of 50% still stands.

Assignment #	Due date	Textbook Topics included
1	Sep 24	1.1,1.2,1.4
2	Sep 24	1.5,2.2
3	Oct 8	2.3
4	Oct 8	2.4, 2.6
5	Oct 22	3.1,3.4
6	Oct 22	3.2,3.3
7	Nov 05	3.5,3.6
8	Nov 05	3.8,3.9
9	Nov 26	4.1,4.3
10	Nov 26	4.4,4.5
11	Dec 11	4.6,4.7,4.10
12	Dec 11	5.1–5.3

Table 1: Knweton Alta Assignment Schedule

8.3 Quizzes

You are required to attend a lab section with this course that will be led by an experienced student mentor. All quizzes happen during lab time. You MUST attend your scheduled lab to get any credit for quizzes.

8.4 Final exam

Final exam information will be released on Aurora when it's near to the end of the semester.

9 Using Copyrighted Materials

Please respect copyright. We may use some copyrighted content in this course. Copyrighted works, including those created by members of the teaching team, are made available for private study and research purpose and must not be distributed in any format without permission. Do not upload copyrighted works to a learning management system (such as UM Learn), or any website, unless an exception to the Copyright Act applies or written permission has been confirmed. For more information, see the University's Copyright Office website at http://umanitoba.ca/copyright/ or contact um_copyright@umanitoba.ca.

10 Recording Class Lectures

Your instructor and the University of Manitoba hold copyright over the course materials, presentations and lectures which form part of this course. No audio or video recording of lectures or presentations is allowed in any format, openly or surreptitiously, in whole or in part without permission. Course materials (both paper and digital) are for the participant's private study and research.

11 Course Technology

We will be using Piazza for after-class discussions. I encourage you to post your questions there. To sign up, click class signup link and join as a student. If asked the term is Fall 2020.

When you email your instructor, please use your University of Manitoba email and include the course code in the title. Your emails will be answered within 2 business days.

12 Student Accessibility Services And Math Help Center

- If you are a student with a disability, please contact SAS for academic accommodation supports and services such as note-taking, interpreting, assistive technology and exam accommodations. Students who have, or think they may have, a disability (e.g. mental illness, learning, medical, hearing, injury-related, visual) are invited to contact SAS to arrange a confidential consultation.

  Student Accessibility Services
  520 University Centre
  204 474 7423
  student_accessibility@umanitoba.ca

- **Math Help Centre**

  The math department has a Math Help Centre, located 311 Machray Hall, available to students registered in first year math courses. You may book one-on-one tutoring service there from Mon to Fri, 9:00AM to 6:00 PM (Jan 16 to Apr 28).

  For details: https://sci.umanitoba.ca/mathematics/undergraduate-info/help-centre/

  Contact person:
  Clifford Allotey
  425 Machray Hall
  E-mail: Clifford.Allotey@umanitoba.ca

13 Academic Integrity

The Faculty of Science and The University of Manitoba regard acts of academic dishonesty in quizzes, tests, examinations, laboratory reports or assignments as serious offences and may assess a variety of penalties depending on the nature of the offence. Acts of academic dishonesty include, but are not limited to bringing unauthorized materials into a test or exam, copying from another individual, using answers provided by tutors, plagiarism, and examination personation. Note: cell phones, pagers, PDAs, MP3 units or electronic translators are explicitly listed as unauthorized materials, and must not be present during tests or examinations unless otherwise stated. Penalties that may apply, as provided for under the University of Manitoba's Student Discipline By-Law, range from a grade of zero for the assignment or examination, failure in the course, to expulsion from the University.

All governing documents may be accessed on the University of Manitoba web-page:The Governing Documents of the University of Manitoba. More information about academic integrity is also available at Academic Integrity, University of Manitoba.

All students are advised to familiarize themselves with the **Student Discipline Bylaw**, which is printed in its entirety in the Student Guide; also available on-line or through the Office of the University Secretary. For more information on suggested minimum penalties assessed by the Faculty of Science for acts of academic dishonesty please visit Faculty of Science Suggested minimum penalties for common acts of academic dishonesty and violation .

14 Notice Regarding Collection, Use, and Disclosure of Personal Information by the University

Your personal information is being collected under the authority of The University of Manitoba Act. It will be used for the purposes of grading papers and providing feedback to students. Personal information will not be used or disclosed for other purposes, unless permitted by The Freedom of Information and Protection of Privacy Act (FIPPA). The University of Manitoba has taken steps to ensure that its agreements with Crowdmark, Inc. and WebAssign for services provided by the Crowdmark and WebAssign applications are in compliance with FIPPA. Please be aware that information held by Crowdmark Inc. and Webassign may be transmitted to and stored on servers outside of the University of Manitoba, or Canada. The University of Manitoba cannot and does not guarantee protection against the possible disclosure of your data including, without limitation, against possible secret disclosures of data to a foreign authority in accordance with the laws of another jurisdiction. If you have any questions about the collection of personal information, contact the Access and Privacy Office (tel. 204-474-9462), The University of Manitoba, 233 Elizabeth Dafoe Library, Winnipeg, Manitoba, Canada, R3T 2N2.

15 Land Acknowledgement

The University of Manitoba campuses are located on original lands of Anishinaabeg, Cree, Oji-Cree, Dakota, and Dene peoples, and on the homeland of the Métis Nation.

We respect the Treaties that were made on these territories, we acknowledge the harms and mistakes of the past, and we dedicate ourselves to move forward in partnership with Indigenous communities in a spirit of reconciliation and collaboration.